2026 OFFICIAL GLOBAL HARDWARE PORTFOLIO & ENGINEERING CATALOG

Human-Centric Bionics. Precision Thermal Armor.

Guangdong Langcheng Technology Co., Ltd. · Intelligent Protection Technology R&D Center · GDUT Institute of CNC Equipment Collaborative Innovation



MASTER 8-MODEL ENGINEERING SPECIFICATION MATRIX

MODEL CODE	PRODUCT NAME	CORE MECHANISM	CAPACITY / PRECISION	NET WEIGHT
LC Power-U/A Doc: LG-DS-2026-01	**Passive Upper Limb Exoskeleton**	Passive Mechanical Carbon + 4-Angle Ratchet	**Max. 8kg Overhead Assist (100% Passive (0V))**	**< 2kg**
LC Flex-W Doc: LG-DS-2026-02	**Flexible Waist Exoskeleton**	Passive Elastic Polyester Latex + Thigh Linkage	**0-30kg Dynamic Lumbar Assist (100% Passive (0V))**	**<= 1kg**
LC Gloves-01 Doc: LG-DS-2026-03	**Wrist Repetitive Work Assessment Glove**	Dual-IMU Differential Sensing + Edge-AI MCU	**<= ±0.1° Static Accuracy (±180° Range) (Real-Time Wireless Telemetry (24/min))**	**200g**
LC Frost-PCM/A Doc: LG-DS-2026-04	**PCM Cooling Vest (Transparent TPU Edition)**	Passive Bio-Based C17 PCM in Transparent TPU	**22.08°C Constant (225 J/g) (1-2h Cooling / 1h Recharge (-18°C))**	**0.55kg**
LC Frost-PCM/G Doc: LG-DS-2026-05	**Flame Retardant PCM Cooling Vest**	Passive Bio-Based C17 PCM in Antistatic FR Shell	**22.08°C Constant (225 J/g) (1-2h Cooling / 0V Intrinsically Safe)**	**0.96kg**
LC Frost-TEC/W Doc: LG-DS-2026-06	**Semiconductor Water Cooling Vest**	Active 80W TEC Liquid Loop + Dual Chest Fans	**80W Active Cooling (Stable in 2 min) (24V / 100W | 2-4h Endurance)**	**< 3kg**
LC Frost-TEC/D Doc: LG-DS-2026-07	**Direct Cooling Semiconductor Vest**	4-Pod Direct-Contact Solid-State Peltier (Dry)	**45W Direct Conduction (2s Instant) (12V / 60W | 2-8h Modular Runtime)**	**~1.7kg**
LC Frost-ACC/W Doc: LG-DS-2026-08	**Compressor Water Cooling Vest**	Active Micro-Inverter Compressor + Closed-Loop Liquid	**60-300W Active Refrigeration (Stable in 3 min) (24V / 110W | 2-4h Continuous)**	**< 4kg**

FLAGSHIP 01 · BIONIC EXOSKELETON SERIES

LC Power-U/A (Spec: LC Power-U/P)

Passive Upper Limb Exoskeleton · Overhead Bionic Armor

Overhead Bionic Armor. Pure Mechanical Relief.

Engineered with a pure mechanical bionic structure for precise directional assistance, the LC Power-U/A is lightweight, highly reliable, and adaptable to complex industrial environments. Featuring multi-level adjustments to fit diverse body types, it is purpose-built for automotive assembly, power maintenance, construction, aircraft repair, and agricultural picking - dramatically reducing shoulder and arm muscle load while elevating operational efficiency and all-day comfort.

01 Featherlight Carbon Frame

Aerospace carbon fiber paired with wear-resistant Oxford cloth delivers unmatched rigidity at sub-2kg weight, ensuring unrestricted arm mobility.

02 Intrinsically Safe Zero-Power

100% mechanical elastic energy storage eliminates battery maintenance, charging downtime, and spark hazards across continuous multi-shift operations.

03 High-Efficiency Musculoskeletal Offloading

Slashing deltoid muscle strain by up to 8kg, directly extending worker endurance and preventing chronic shoulder disorders.

04 Minimalist Industrial Durability

High-strength anodized aluminum joints withstand rigorous impacts, dust, and continuous industrial wear.

LC Power-U/A · OFFICIAL HARDWARE VIEW



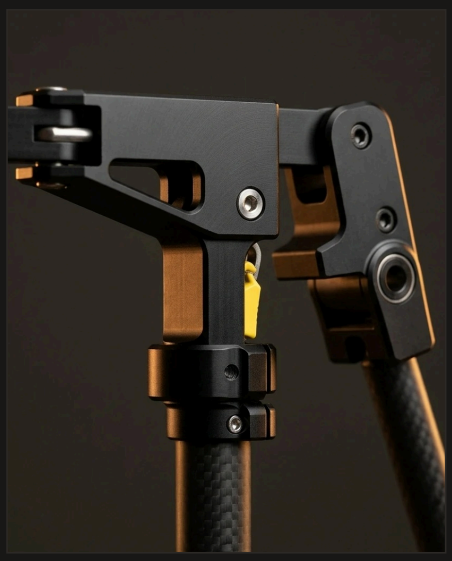
KEY METRIC 01
8kg
Maximum Overhead Assist Capacity

KEY METRIC 02
< 2kg
Ultra-Light Aerospace Carbon Fiber Spine

KEY METRIC 03
100%
Passive Zero-Power Operation (Zero Battery Risk)

KEY METRIC 04
4 Angles
Multi-Gear Articulation (50°, 70°, 90°, 110°/115°)

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



AVIATION ALUMINUM ALLOY · 4-ANGLE LOCK

4-Angle Semi-Automatic Mechanical Shoulder Joint

CNC-machined aviation aluminum alloy housing encloses a 3-gear, 4-angle (50°, 70°, 90°, 110°/115°) semi-automatic locking ratchet that engages silently as the operator raises their arms overhead and disengages smoothly on descent.



T700 AEROSPACE CARBON FIBER

Aerospace Carbon Fiber Telescoping Spine

Multi-directional woven aerospace carbon fiber spine resists torsional flex while transferring up to 8kg of overhead tool weight down to the pelvic belt.



OXFORD OUTER + 3D HONEYCOMB MESH

Quick-Release Armrest Cuffs & 3D Air-Mesh

Engineered with tear-resistant industrial Oxford outer fabric and 3D honeycomb air-mesh inner lining for rapid emergency release and 8-hour thermal ventilation.

ANATOMICAL & MECHANICAL SUBSYSTEMS (5 NODES)

1. **4-Gear Shoulder Joint [Glenohumeral Articulation Unit]**
   Semi-automatic mechanical assist engages instantly as arms elevate, with locks at 50°, 70°, 90°, 110°/115°.
2. **Quick-Release Chest & Dorsal Harness [Thoracic Cross-Lock Harness]**
   Ergonomic snap buckles enable rapid donning and doffing in under 15 seconds.
3. **Telescoping Carbon Back Bar [Dorsal Carbon Spine]**
   High-tensile carbon fiber back spine seamlessly adjusts to different torso lengths.
4. **Quick-Release Armrest Cuffs [Humeral Support Cradle]**
   Cuffed ergonomic armrests with yellow buckle and blue quick-release tab provide immediate escape.
5. **Contoured 3D Lumbar Belt [Pelvic Load-Transfer Girdle]**
   Breathable air-mesh waist belt distributes harness load evenly across the pelvic girdle.

FIELD DEPLOYMENT | AUTOMOTIVE FINAL ASSEMBLY LINE | OVERHEAD CHASSIS STATION



Underbody Automotive Assembly & Overhead Torque Operations

Inside modern automotive assembly plants, technicians execute thousands of overhead fastening cycles per shift beneath elevated vehicle chassis. The LC Power-U/A locks in mechanical support at 50° to 115°, neutralizing the weight of pneumatic torque wrenches and eliminating deltoid fatigue without requiring a single watt of electrical power.

Up to 8kg Offload	< 15 Seconds	0 Hours (100% Passive)
Deltoid EMG Reduction	Donning / Doffing Time	Power & Charging Downtime

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Power-U/A (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Power-U/A (Spec Code: LC Power-U/P)
IDENTITY & COMPLIANCE	**Color**	Industrial Matte Black
IDENTITY & COMPLIANCE	**Target Users**	Workers with prolonged arm-raising and overhead tasks
IDENTITY & COMPLIANCE	**Applications**	Automotive assembly, power maintenance, construction, aircraft repair, agricultural harvesting
MATERIALS & ERGONOMICS	**Structural Materials**	Carbon fiber + aviation aluminum alloy
PERFORMANCE & MECHANICS	**Assistance Capacity**	Max. 8kg
PERFORMANCE & MECHANICS	**Locking Mode**	Semi-automatic mechanical locking
PERFORMANCE & MECHANICS	**Assistance Gears**	3 gears (4-angle engagement)
PERFORMANCE & MECHANICS	**Assistance / Unlocking Angles**	50°, 70°, 90°, 110°/115°
MATERIALS & ERGONOMICS	**Outer Fabric**	Wear-resistant Oxford fabric
MATERIALS & ERGONOMICS	**Lining Fabric**	Breathable 3D mesh fabric
DIMENSIONS & WARRANTY	**Sizes**	Standard Size / Plus Size
DIMENSIONS & WARRANTY	**Net Weight**	< 2kg
DIMENSIONS & WARRANTY	**Package Dimensions**	50 x 40 x 12cm
DIMENSIONS & WARRANTY	**Optional Accessories**	Ergonomic neck support collar
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Power-U/A

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware

FLAGSHIP 02 · BIONIC EXOSKELETON SERIES

LC Flex-W

Flexible Waist Exoskeleton · Unpowered Smart Lumbar Belt

The Unpowered Smart Belt. Up to 30kg Dynamic Lumbar Thrust.

This lightweight, unpowered lumbar exoskeleton is designed for frequent bending and lifting tasks. It stores and releases energy via elastic material deformation, acting as a biomechanical "smart belt" to deliver upward assistive thrust during bending. Significantly reduces lumbar strain, offloads spinal disc compression, and mitigates occupational lower-back injury risks.

01 Sub-1kg Minimalist Chassis

Breathable textile construction conforms naturally to the torso, guaranteeing sweat-free comfort throughout an 8-hour shift.

02 Unrestricted Agility

Conforms to body contours without rigid metal hip constraints, preserving full freedom for walking, crouching, climbing, and driving.

03 Dynamic Erector Spinae Offloading

Progressively builds elastic tension as the user bends forward, releasing explosive upward thrust upon standing.

04 Spinal Pressure Equilibrium

Distributes acute lumbar point loads across the thoracic spine and thighs, preventing occupational disc herniation.

LC Flex-W · OFFICIAL HARDWARE VIEW



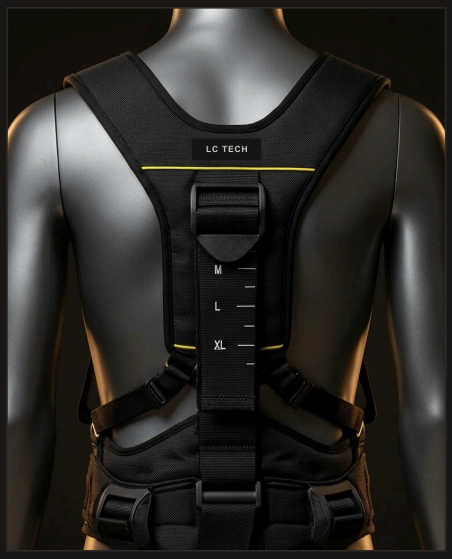
KEY METRIC 01
0-30kg
Dynamic Upward Lumbar Assist Range

KEY METRIC 02
<= 1kg
Ultra-Slim Wearable Profile

KEY METRIC 03
160-190cm
Broad Ergonomic Height Adaptation

KEY METRIC 04
0V
Non-Powered Fail-Safe Kinetic Energy Storage

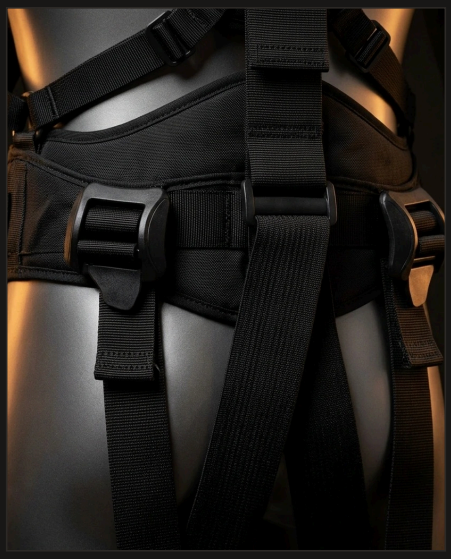
MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



3D BREATHABLE MESH + LUMBAR BRACKET

Integrated Back-Lumbar Support Bracket & 3D Cushion

An ergonomic multi-layer 3D honeycomb lumbar cushion houses the integrated back-lumbar support bracket, stabilizing the lower spine while venting heat and moisture during high-frequency warehouse sorting.



HIGH-RESILIENCE POLYESTER LATEX

High-Resilience Polyester Latex Fiber Core

Engineered elastic polyester latex fibers store mechanical potential energy during forward trunk flexion and return up to 30kg of smooth upward assistive thrust.



WEAR-RESISTANT TACTICAL OXFORD

Bilateral Thigh Linkage Strap System

Soft wear-resistant Oxford thigh cuffs anchor the kinetic load pathway without rigid metal hip hinges, allowing operators to transition directly from lifting to driving forklifts.

ANATOMICAL & MECHANICAL SUBSYSTEMS (5 NODES)

1. **Multi-Height Adjustable Back [Thoracic Tension Anchor]**
   Telescoping shoulder tension system accommodates operator heights from 160cm to 190cm.
2. **Flank Articulation Routing [Lateral Oblique Guide]**
   Soft flexible strap pathways eliminate hip pinching during extreme deep squats.
3. **Thickened 3D Lumbar Cushion [L1-L5 Decompression Pad]**
   Ergonomic multi-layer pad cushions the lower spine and distributes back loads.
4. **Fail-Safe Elastic Core [Kinetic Deformation Band]**
   High-resilience polyester latex fibers ensure lifetime durability in wet or dusty workplaces.
5. **Bilateral Thigh Traction Straps [Femoral Linkage Cuffs]**
   Integrated lumbar support bracket linked to bilateral thigh cuffs for smooth 0-30kg load transfer.

FIELD DEPLOYMENT | SMART LOGISTICS FULFILLMENT CENTER | PALLETIZING ZONE



High-Frequency Logistics Hubs & Airport Baggage Handling

In high-throughput logistics sorting centers and airport baggage bays, handlers perform hundreds of deep bends and heavy lifts every hour. Weighing <=1kg with zero rigid hip bars, the LC Flex-W acts as a second set of erector spinae muscles - delivering up to 30kg of upward lumbar assist while permitting natural walking, climbing, and vehicle operation.

0-30kg Adjustable	<= 1.0kg	160-190cm
Dynamic Upward Assist	Total Wearable Weight	Operator Height Range

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Flex-W (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Flex-W
IDENTITY & COMPLIANCE	**Color**	Tactical Black
IDENTITY & COMPLIANCE	**Target Users**	Manual laborers, warehouse operators, logistics handlers, caregivers
IDENTITY & COMPLIANCE	**Applications**	Industrial logistics, airport baggage, medical elderly care, frequent bending operations
PERFORMANCE & MECHANICS	**Assistance Type**	Non-powered passive mechanical
DIMENSIONS & WARRANTY	**Applicable Height**	160-190cm
PERFORMANCE & MECHANICS	**Assistance Range**	0-30kg (dynamically adjustable)
MATERIALS & ERGONOMICS	**Energy Storage Medium**	High-resilience elastic polyester latex fiber
PERFORMANCE & MECHANICS	**Core Structure**	Integrated back-lumbar support bracket + bilateral thigh linkage strap system
MATERIALS & ERGONOMICS	**Outer Fabric**	Wear-resistant Oxford fabric
MATERIALS & ERGONOMICS	**Lining Fabric**	Breathable 3D mesh fabric
DIMENSIONS & WARRANTY	**Sizes**	Standard Size / Plus Size
DIMENSIONS & WARRANTY	**Net Weight**	<= 1kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Flex-W

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware

FLAGSHIP 03 · SMART ERGONOMIC SENSING SERIES

LC Gloves-01

Wrist Repetitive Work Assessment Glove · Edge-AI Biometric Diagnostic Glov

Edge-AI Biometric Glove. GBZ 336-2025 Carpal Tunnel Risk Diagnostics.

Designed for manufacturing jobs with repetitive forceful wrist movements, the LC Gloves-01 strictly complies with GBZ 336-2025 Diagnostic Criteria for Occupational Carpal Tunnel Syndrome. Equipped with dual-posture IMU sensors (dorsal hand + wrist cuff) and onboard edge-AI computing, it tracks wrist angles, valid work motions, operation frequency, duration, and load intensity in real time. Supports wireless data upload and tablet analytics for ergonomic risk quantification and workstation optimization.

01 Dual-Point Attitude Sensing

Differential sensing architecture cancels whole-arm movement artifacts, isolating pure wrist flexion, extension, and deviation.

02 Onboard Edge-AI Processing

Glove micro-controller calculates angles, counts valid cycles, and compiles metrics locally with zero latency.

03 Wireless Tablet Telemetry

Real-time Bluetooth/Wi-Fi data stream to companion PC/tablet suite generates exportable diagnostic risk reports.

04 Industrial-Grade Anti-Interference

Shielded against electromagnetic noise and machine vibration in automotive and electronics plants.

LC Gloves-01 · OFFICIAL HARDWARE VIEW



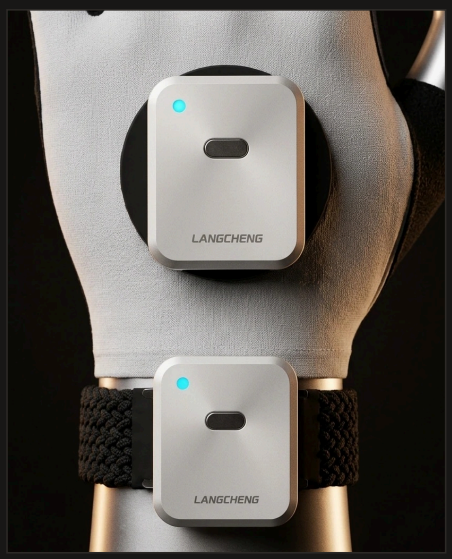
KEY METRIC 01
<= ±0.1°
Static IMU Measurement Precision

KEY METRIC 02
±180°
Full Multi-Axis Range of Motion

KEY METRIC 03
200g
Featherlight Ergonomic Form Factor

KEY METRIC 04
24/min
Automated Valid Motion Frequency Tracking

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



DUAL DIFFERENTIAL IMU · · ±0.1° PRECISION

Dual-Posture Differential IMU Sensor Pods

By pairing a dorsal metacarpal IMU with a wrist-cuff reference IMU, the onboard edge-AI processor rejects gross arm motion and isolates true wrist flexion/extension and radial/ulnar deviation past the 15° recognition threshold.



TECHNICAL GREY ANTI-SLIP TEXTILE

High-Stretch Technical Knit & Half-Finger Cutouts

200g breathable technical grey textile keeps IMU sensors locked without skin migration while leaving fingertips free for micro-fasteners and soldering.



GBZ 336-2025 COMPLIANT ANALYTICS

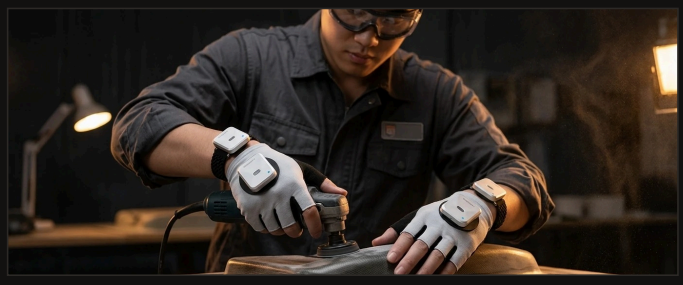
GBZ 336-2025 Diagnostic Waveform Analytics Suite

Streams real-time wrist kinematics to an industrial tablet PC to log daily working hours, valid motion counts, and automated occupational carpal tunnel syndrome risk reports.

ANATOMICAL & MECHANICAL SUBSYSTEMS (5 NODES)

1. **Half-Finger Tactile Cutouts [Distal Phalangeal Cutouts]**
   Exposed fingertips preserve fine touch for delicate assembly of miniature screws and electronics.
2. **Zero-Slack Dorsal Pod Mount [Primary Metacarpal IMU Node]**
   High-stretch elastic fabric locks the primary IMU sensor firmly over the metacarpals.
3. **Lightweight 200g Architecture [Breathable Technical Knit Chassis]**
   Minimizes muscle load during continuous 8-hour ergonomic audits.
4. **Adjustable Secondary Wristband [Radiocarpal Reference Strap]**
   Anchors the reference IMU sensor across varied wrist circumferences.
5. **Companion Diagnostic Software & Reference IMU [Wireless Edge-AI Telemetry Link]**
   Secondary wrist IMU pod streams real-time dual-channel waveforms for flexion/extension, radial/ulnar deviation, and GBZ 336-2025 risk grading.

FIELD DEPLOYMENT | 3C ELECTRONICS MANUFACTURING LINE | ERGONOMIC AUDIT STATION



Precision Electronics Assembly & Occupational Health Audits

Occupational disease prevention hospitals, third-party ergonomic testing agencies, and manufacturing EHS teams deploy the LC Gloves-01 directly on live production lines. As operators assemble precision components, the glove captures every flexion, extension, and ulnar deviation above the customizable 15° threshold - delivering legally defensible GBZ 336-2025 compliance reports.

GBZ 336-2025 Compliant	<= ±0.1° (±180° Range)	15° / 24 Motions min/min
Diagnostic Standard	Static Angle Accuracy	Standard Motion Threshold

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Gloves-01 (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Gloves-01
IDENTITY & COMPLIANCE	**Color**	Technical Light Grey with Matte White Sensor Pods & Black Trim
IDENTITY & COMPLIANCE	**Compliance**	GBZ 336-2025 Diagnostic Criteria for Occupational Carpal Tunnel Syndrome
IDENTITY & COMPLIANCE	**Applications**	Occupational disease prevention hospitals, third-party testing agencies, manufacturing workstation audits
PERFORMANCE & MECHANICS	**Recognition Threshold**	15° (customizable)
PERFORMANCE & MECHANICS	**Angle Range**	±180°
PERFORMANCE & MECHANICS	**Static Accuracy**	<= ±0.1°
PERFORMANCE & MECHANICS	**Frequency Counting**	Automatically counts valid motions (standard: 24 motions/min)
PERFORMANCE & MECHANICS	**Duration Tracking**	Daily working hour accumulation
PERFORMANCE & MECHANICS	**Measurement Dimensions**	Wrist flexion/extension, radial/ulnar deviation angles
DIMENSIONS & WARRANTY	**Sizes**	M, L, XL (customizable)
DIMENSIONS & WARRANTY	**Net Weight**	200g
DIMENSIONS & WARRANTY	**Package Dimensions**	23 x 17 x 8cm
DIMENSIONS & WARRANTY	**Optional Accessories**	Industrial Tablet PC (with pre-installed analytics suite) / Action Camera
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Gloves-01

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware

FLAGSHIP 04 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-PCM/A

PCM Cooling Vest (Transparent TPU Edition) · Bio-Based C17 Phase Change

Bio-Based Phase Change Armor. Constant 22.08°C Torso Comfort.

Built with high-energy-density Phase Change Material (PCM) that absorbs body heat at a constant comfort temperature (~23°C), the LC Frost-PCM/A delivers gentle, long-lasting torso cooling. A single cold-storage cycle provides 1-2 hours of relief in 35°C high-heat environments. Reusable thousands of times - an intrinsically safe, wire-free personal micro-climate.



01 Gentle Human-Comfort Cooling

Operates at 22.08°C, completely preventing skin vasoconstriction or frostbite risks caused by traditional 0°C ice packs.

02 100% Passive & Intrinsically Safe

Requires no batteries, fans, or wires. Zero electrical hazards in cleanrooms, hazardous gas zones, and wet areas.

03 Zero-Degradation C17 Formulation

Pure bio-based C17 n-alkane PCM withstands thousands of freeze-thaw cycles without performance loss.

04 Plant-Derived & Biodegradable

100% biomass formulation, non-petroleum derived, environmentally safe and non-toxic.

KEY METRIC 01
22.08°C
Constant Phase Change Comfort Temperature

KEY METRIC 02
225 J/g
High Latent Heat of Fusion

KEY METRIC 03
0.55kg
Ultra-Lightweight Transparent TPU Chassis

KEY METRIC 04
1-2 Hours
Continuous Cooling per Charge Cycle

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



100% BIO-BASED C17 PCM · 225 J/G

Bio-Based C17 n-Alkane Phase Change Matrix

Plant-derived C17 phase change crystals transition between solid and liquid at an exact 22.08°C plateau, absorbing 225 J/g of metabolic heat without causing cold shock or condensation dampness.



HIGH-STRENGTH TRANSPARENT TPU

High-Strength Medical-Grade TPU RF Welded Seams

High-frequency welded Thermoplastic Polyurethane (TPU) grid channels resist puncture, chemical splashes, and biological contamination for sterile medical and cleanroom deployment.



DUAL SIDE-RELEASE TACTICAL BUCKLES

Universal Dual Side-Release Webbing System

Low-profile acetal buckles and heavy-duty webbing secure the 0.55kg vest beneath sealed hazmat or cleanroom coveralls with zero bulk.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **Fast-Cooling Grid Matrix [Anterior & Dorsal C17 Cell Matrix]**
   Independent encapsulated PCM cells conform across the chest and back for maximum heat exchange area.

2. **Universal Adjustable Buckles [Lateral Tension Fasteners]**
   Dual side-release black webbing buckles adapt to all body sizes and uniform styles.

3. **Medical-Grade TPU Shell [High-Frequency Welded TPU Encapsulation]**
   Waterproof and oil-resistant transparent film enables instant sanitization and wipe-down cleaning.

4. **Low-Profile Under-Layer Cut [0.55kg Sub-Suit Ergonomic Cut]**
   Ultra-slim 0.55kg profile fits seamlessly under hazmat suits, cleanroom suits, or high-vis vests.

FIELD DEPLOYMENT | SEMICONDUCTOR CLEANROOM & BIO-SAFETY LEVEL PPE DEPLOYMENT



Sealed Cleanrooms, Medical PPE & High-Heat Sanitation Operations

Operators wearing impermeable hazmat suits, medical PPE coveralls, or cleanroom bunny suits cannot rely on evaporative sweat cooling or fan-blown air. At just 0.55kg, the crystal-transparent LC Frost-PCM/A slips cleanly beneath sealed outer garments, maintaining a constant 22.08°C micro-climate for 1-2 hours per cycle and recharging in 60 minutes at -18°C.

22.08°C Constant	225 J/g	1 Hour at -18°C
Phase Change Plateau	Latent Heat Capacity	Fast Freezer Recharge

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-PCM/A (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-PCM/A
IDENTITY & COMPLIANCE	**Color**	Crystal Transparent (customizable PCM dye options)
IDENTITY & COMPLIANCE	**Target Users**	Workers in extreme high-temperature environments, cleanrooms, medical PPE, sanitation
PERFORMANCE & MECHANICS	**Heat of Fusion**	225 J/g
PERFORMANCE & MECHANICS	**Phase Change Temperature**	22.08°C
PERFORMANCE & MECHANICS	**Cooling Duration**	1-2 hours (reference based on labor intensity and ambient temperature)
PERFORMANCE & MECHANICS	**Cold Storage Recharge Time**	Frozen at -18°C for 1 hour, OR refrigerated at 0-4°C for 1-2 hours
MATERIALS & ERGONOMICS	**Shell Material**	High-strength Thermoplastic Polyurethane (TPU)
DIMENSIONS & WARRANTY	**Size**	One Size Fits All
DIMENSIONS & WARRANTY	**Net Weight**	0.55kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Optional Accessories**	Custom PCM dye colors
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-PCM/A

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware

FLAGSHIP 05 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-PCM/G

Flame Retardant PCM Cooling Vest · Antistatic & Flame-Retardant PCM Armo

Antistatic & Flame-Retardant PCM Armor. Passive Cooling for Hazardous Frontlines.

Engineered with high-energy-density Phase Change Material absorbing heat at constant ~23°C and encapsulated within an antistatic, flame-retardant outer shell. Designed for petrochemical, foundry, welding, and utility operations where spark hazards, explosive dust, and extreme heat converge.

01 Flame-Retardant & Antistatic Protection

Heavy-duty industrial twill fabric repels slag, sparks, and static charge accumulation.

02 Constant 22.08°C Thermal Regulation

Maintains continuous core cooling without cold shock during heavy manual labor in 35°C+ ambient heat.

03 High-Stability C17 Biomass Core

Thousands of thermal cycles without degradation; non-toxic and eco-friendly.

04 Night & Low-Light Visibility

Dual high-visibility silver reflective bands provide 360° safety in roadworks and dark industrial plants.

LC Frost-PCM/G · OFFICIAL HARDWARE VIEW



KEY METRIC 01
22.08°C
Non-Irritating Phase Change Comfort Zone

KEY METRIC 02
225 J/g
Latent Heat Storage Capacity

KEY METRIC 03
360°
High-Visibility Retro-Reflective Safety Bands

KEY METRIC 04
0V
Zero-Spark Intrinsically Safe Passive Cooling

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



FR & ANTISTATIC INDUSTRIAL TWILL

Antistatic & Flame-Retardant Ripstop Outer Shell

Specially treated antistatic and flame-retardant industrial weave dissipates electrostatic charges and resists welding splatter, enabling safe core cooling inside petrochemical refineries and gas plants.



360° RETRO-REFLECTIVE TRIM

Dual 360° Retro-Reflective Safety Warning Bands

Wide silver retro-reflective bands wrap 360° around the chest and waist, ensuring immediate operator recognition in low-light refineries, tunnels, and night-shift roadworks.



MODULAR PCM BAYS · 0.96KG TOTAL

Modular Internal PCM Bays & Elastic Waistband

Internal thermal compartments allow crews to swap freshly recharged 22.08°C C17 PCM packs in seconds while the wide magic-tape waistband locks the vest closely to the torso.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **FR & Antistatic Outer Weave [Flame-Retardant Industrial Twill Shell]**
   Rugged ripstop fabric resists tearing, abrasions, and molten metal splatter.
2. **Elastic Magic-Tape Waistband [Heavy-Duty Cummerbund Fastener]**
   Wide industrial hook-and-loop cummerbund fits securely across all torso builds.
3. **Modular PCM Thermal Pockets [Internal C17 Thermal Pack Bays]**
   Dedicated internal compartments hold sealed PCM packs tight against the core.
4. **Dual Safety Warning Reflective Bands [360° Retro-Reflective Warning Strips]**
   3M-grade retro-reflective stripes across the chest and waist ensure field visibility.

FIELD DEPLOYMENT | EX-RATED PETROCHEMICAL CRACKING UNIT | HIGH-HEAT ZONE



Petrochemical Refineries, Welding Bays & Night-Shift Utilities

In explosion-proof (EX) petrochemical zones and active welding shops, electronic batteries and brushed motors pose unacceptable ignition risks. The LC Frost-PCM/G pairs 0V passive 22.08°C phase-change cooling with an antistatic, flame-retardant outer shell and 360° reflective bands - delivering uncompromising thermal relief where electronics are forbidden.

0V Intrinsically Safe	22.08°C (225 J/g)	0.96kg
Ignition & Spark Hazard	Phase Change Comfort	Complete Vest Weight

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-PCM/G (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-PCM/G
IDENTITY & COMPLIANCE	**Color**	Industrial Grey with High-Visibility Reflective Trim (customizable)
IDENTITY & COMPLIANCE	**Target Users**	Petrochemical, metallurgy, welding, power grid, and high-temperature night-shift workers
PERFORMANCE & MECHANICS	**Heat of Fusion**	225 J/g
PERFORMANCE & MECHANICS	**Phase Change Temperature**	22.08°C
PERFORMANCE & MECHANICS	**Cooling Duration**	1-2 hours (reference based on labor intensity and ambient temp)
PERFORMANCE & MECHANICS	**Cold Storage Recharge Time**	Frozen at -18°C for 1 hour, OR refrigerated at 0-4°C for 1-2 hours
MATERIALS & ERGONOMICS	**Fabric Material**	Antistatic and flame-retardant fabric
DIMENSIONS & WARRANTY	**Sizes**	Standard Size / Plus Size
DIMENSIONS & WARRANTY	**Net Weight**	0.96kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-PCM/G

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware

FLAGSHIP 06 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-TEC/W

Semiconductor Water Cooling Vest · 80W Active Liquid + Dual Airflow Chiller

80W Active Thermoelectric Chiller. Closed-Loop Liquid + Dual Airflow.

An active wearable cooling device powered by solid-state thermoelectric cooling (TEC) chips and a closed-loop water circulation system. Delivers steady, multi-level adjustable torso cooling within 2 minutes of startup. Integrates a quiet rear turbine exhaust fan and dual chest air-supply fans for all-day facial and torso dryness in firefighting, construction, power maintenance, and sanitation.

LC Frost-TEC/W · OFFICIAL HARDWARE VIEW



01 Precision Active Temperature Regulation

Thermoelectric chiller actively draws heat away from circulating liquid, overcoming passive melting limitations.

02 Dual-Zone Forced-Air Evaporation

Rear turbine expels heat while dual front fans direct air toward the neck and face to accelerate sweat evaporation.

03 Extended Field Battery Life

24V lithium power supply delivers 1-2 hours on a single pack and up to 4 hours with dual batteries.

04 Leak-Proof Sealed Liquid Channels

Industrial TPU tubing resists high pressures, twisting, and rough site conditions without leaking.

KEY METRIC 01
80W
Active Thermoelectric Cooling Capacity

KEY METRIC 02
2 min
Fast Thermal Stabilization Time

KEY METRIC 03
24V
High-Efficiency Chiller Architecture

KEY METRIC 04
2-4 Hours
Continuous Shift Battery Endurance

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



80W TEC CHILLER · 2-MIN STABILIZATION

Dorsal 80W Solid-State TEC Chiller & Turbine Exhaust

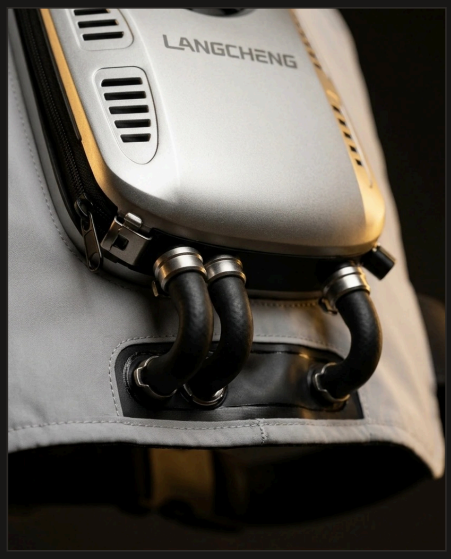
Mounted ergonomically on the back, the silver-anodized 80W thermoelectric module chills internal circulating water within 2 minutes while a quiet radial turbine exhausts rejected heat away from the operator.



DUAL ANTERIOR UPDRAFT FANS

Dual Chest Forced-Air Supply Fans

Integrated front pocket blowers channel fresh airflow upward across the chest, neck, and face, combining internal conductive liquid cooling with external convective drying.



CLOSED-LOOP TPU FLUIDIC LINER

Medical/Industrial TPU Closed-Loop Water Liner

Serpentine TPU micro-channels distribute chilled water evenly across the core and may be paired with supplemental PCM cooling packs for extreme thermal spikes.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **Antistatic FR V-Neck Shell [Anterior Flame-Retardant Chassis]**
   Soft, protective industrial grey fabric resistant to heat and static hazards.

2. **Side Compression Straps [Bilateral Thermal Contact Tensioners]**
   Ergonomic tensioners pull cooling water channels tight against the torso.

3. **Rear 80W TEC Chiller Module [Dorsal 24V / 80W Thermoelectric Core]**
   Sleek silver casing with black turbine fan stabilizes water temperatures within 2 minutes.

4. **Dual Front Air-Supply Pockets [Anterior Updraft Air-Supply Modules]**
   Front utility pockets hold batteries while driving cooling airflow up toward the chin and chest.

FIELD DEPLOYMENT | 500KV SUBSTATION TRANSFORMER OVERHAUL | SUMMER PEAK LOAD



High-Voltage Power Grid Maintenance, Firefighting & Heavy Construction

When power grid crews and firefighters operate in 40°C+ radiant heat, passive ice packs can melt before a complex maintenance task finishes. The LC Frost-TEC/W combines 80W of active thermoelectric liquid chilling with dual chest fans that sweep cooling airflow across the neck and face - sustaining peak cognitive alertness for 2 to 4 hours.

80W (24V / 100W Rated)	2 Minutes	2-4 Hours (Dual Battery)
Active Cooling Output	Loop Stabilization	Continuous Endurance

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-TEC/W (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-TEC/W
IDENTITY & COMPLIANCE	**Color**	Technical Grey (customizable)
IDENTITY & COMPLIANCE	**Target Users**	Firefighting, public security, power grid, construction, sanitation in extreme heat
PERFORMANCE & MECHANICS	**Cooling Capacity**	80W
PERFORMANCE & MECHANICS	**Rated Voltage**	24V
PERFORMANCE & MECHANICS	**Rated Power**	100W
PERFORMANCE & MECHANICS	**Cooling Response Time**	Stable in 2 min
PERFORMANCE & MECHANICS	**Battery Life**	1-2 hours single battery (2-4 hours continuous system endurance)
MATERIALS & ERGONOMICS	**Outer Fabric**	Antistatic and flame-retardant fabric
MATERIALS & ERGONOMICS	**Lining Fabric**	Medical/industrial TPU water-channel liner
DIMENSIONS & WARRANTY	**Size**	One Size Fits All
DIMENSIONS & WARRANTY	**Net Weight**	< 3kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Optional Accessories**	Supplemental PCM cooling packs
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-TEC/W

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware

FLAGSHIP 07 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-TEC/D

Direct Cooling Semiconductor Vest · 2-Second Instant Solid-State Peltier Vest

2-Second Instant Peltier Cooling. Zero Water, Zero Refrigerant, 8-Hour Endurance.

Leveraging the solid-state Peltier effect via direct-contact TEC modules, the LC Frost-TEC/D requires no ice, liquid coolant, or compressor. Direct-contact thermal conduction plates absorb torso heat immediately upon power-on - stabilizing in just 2 seconds - while whisper-quiet fans dissipate waste heat. Features a breathable mesh chassis and high-visibility reflective stripes for industrial inspection, medical PPE, and outdoor operations.

01 Instant 2-Second Solid-State Conduction

Direct Peltier contact plates cool immediately upon activation with zero pre-freezing or water refilling.

02 Whisper-Quiet Vibration-Free Operation

Low-noise axial cooling fans eliminate compressor drone and pump vibrations for quiet inspections.

03 Up to 8-Hour Modular Power

Low 60W power draw enables 2-8 hours of runtime (2 hours per lightweight battery pack).

04 Zero Maintenance & 20,000-Hour Longevity

Solid-state architecture eliminates hoses, refrigerants, and pumps for leak-proof durability.

LC Frost-TEC/D · OFFICIAL HARDWARE VIEW



KEY METRIC 01
2 Seconds
Ultra-Fast Solid-State Cooling Stabilization

KEY METRIC 02
0 Liquid
Completely Dry Direct Conduction (Zero Leakage Risk)

KEY METRIC 03
2-8 Hours
Extended Modular Battery Operation

KEY METRIC 04
> 20,000 Hours
Solid-State Semiconductor Lifespan

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



4-POD SOLID-STATE PELTIER · >20,000H LIFESPAN

4-Pod Direct-Contact Solid-State Peltier Conduction Array

Four anodized aluminum thermal conduction plates coupled directly to solid-state semiconductor chips pull heat from the upper back within 2 seconds of power-on - zero water, zero pumps, and >20,000 hours of chip lifespan.



HI-VIS QUICK-RELEASE FASTENERS

Quick-Release Hi-Vis Chest Buckles & Elastic Bungee Tension

High-visibility yellow quick-release buckles and rear dynamic bungee lacing maintain optimal contact pressure between the 4 Peltier pods and the operator's upper back.



DETACHABLE WASHABLE ANTISTATIC FR MESH

Chest Rotary Controller & Modular Washable Mesh

Intuitive chest-mounted rotary dial adjusts solid-state cooling intensity on the fly, while the entire electronic harness detaches so the antistatic FR mesh vest can be washed.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **4-Pod Direct-Contact TEC Array [Quad Dorsal Solid-State Peltier Array]**
   Four Peltier conduction plates positioned on the upper back deliver instantaneous cooling directly to the skin.
2. **Quick-Release Yellow Chest Buckles [Hi-Vis Anterior Fastening Harness]**
   High-visibility chest fasteners and rear bungee cord ensure a snug, non-restrictive fit.
3. **Chest Controller & Low-Noise Fans [Rotary Thermal Dial & Axial Heat Sinks]**
   Chest-mounted circular dial provides intuitive multi-stage thermal regulation.
4. **Detachable Washable Hi-Vis Mesh [Modular Washable Antistatic Mesh Chassis]**
   Breathable antistatic mesh with neon-yellow and silver reflective stripes; electronics detach easily for washing.

FIELD DEPLOYMENT | HIGH-SPEED RAIL CORRIDOR INSPECTION | 8-HOUR PATROL SHIFT



Railway Track Inspection, Outdoor Security & Mobile Field Patrols

Mobile inspection engineers, traffic officers, and outdoor security personnel need immediate cooling without carrying water reservoirs or waiting for freezer packs. Weighing ~1.7kg and stabilizing in just 2 seconds on a 12V battery system, the LC Frost-TEC/D delivers up to 8 hours of dry, whisper-quiet solid-state cooling across long field patrols.

2 Seconds (Instant)	2-8 Hours (12V / 60W)	> 20,000 Hours
Thermal Stabilization	Modular Battery Runtime	Semiconductor Lifespan

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-TEC/D (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-TEC/D
IDENTITY & COMPLIANCE	**Color**	Tactical Grey Mesh with Neon Yellow Hi-Vis Reflective Trim & Black Straps (Black Option)
IDENTITY & COMPLIANCE	**Target Users**	Industrial inspection, medical PPE, outdoor traffic, security personnel
PERFORMANCE & MECHANICS	**Cooling Capacity**	45W
PERFORMANCE & MECHANICS	**Rated Voltage**	12V
PERFORMANCE & MECHANICS	**Rated Power**	60W
PERFORMANCE & MECHANICS	**Cooling Response Time**	Instant cooling, stable in 2s
PERFORMANCE & MECHANICS	**Battery Life**	2-8 hours (2 hours per single battery pack)
PERFORMANCE & MECHANICS	**Chip Lifespan**	> 20,000 hours
MATERIALS & ERGONOMICS	**Outer Fabric**	Antistatic and flame-retardant breathable mesh (modular detachable & washable)
DIMENSIONS & WARRANTY	**Sizes**	Standard Size / Plus Size
DIMENSIONS & WARRANTY	**Net Weight**	~1.7kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-TEC/D

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware

FLAGSHIP 08 · PERSONAL THERMAL MANAGEMENT SERIES

LC Frost-ACC/W

Compressor Water Cooling Vest · 300W Extreme Micro-Inverter Chiller

300W Active Micro-Inverter Refrigeration for Extreme Heat Hazards.

The flagship of the LC Frost line, the LC Frost-ACC/W integrates a miniature inverter refrigeration compressor with a closed-loop chilled liquid circulation network. Delivering an industry-leading 60-300W of active cooling capacity, a wide temperature adjustment range, and unwavering constant-temperature output, it is the ultimate protective cooling armor for extreme metallurgy, firefighting, and confined-space operations.

01 300W Extreme Active Refrigeration

Micro-inverter compressor rapidly chills circulating water upon startup with multi-stage digital temperature regulation.

02 Under-4kg Integrated Mobility

Compact cooling architecture balances weight along the lower torso for unrestricted mobility in tight quarters.

03 Intensive 2-4 Hour Shift Endurance

High-capacity 24V lithium battery system sustains heavy-duty refrigeration across grueling shifts.

04 Shockproof & Sealed Fluidics

Rugged liquid couplings and pressurized TPU bladders withstand heavy impacts, vibration, and site washdowns.

LC Frost-ACC/W · OFFICIAL HARDWARE VIEW



KEY METRIC 01
60-300W
Heavy-Duty Active Compressor Refrigeration

KEY METRIC 02
< 4kg
Ergonomically Balanced Wearable Chiller Unit

KEY METRIC 03
3 min
Rapid High-Capacity Chilled Loop Stabilization

KEY METRIC 04
100%
Sealed Leak-Proof Liquid Circuit

MATERIAL SCIENCE & MICRO-ARCHITECTURE · STRUCTURAL CRAFTSMANSHIP DETAILS



60·300W MICRO-INVERTER COMPRESSOR

24V Miniature Inverter Refrigeration Compressor Core

Unlike standard thermoelectric coolers, the LC Frost-ACC/W utilizes a true vapor-compression micro-inverter cycle capable of delivering up to 300W of refrigeration power - maintaining cold liquid circulation even in 50°C+ foundry radiation.



PRESSURIZED TPU LIQUID BLADDER

High-Durability Pressurized TPU Liquid Cooling Bladder

High-density serpentine TPU cooling bladder maximizes torso contact surface area while an inner breathable 3D mesh barrier prevents skin clamminess.



ANTISTATIC FR SHELL · < 4KG TOTAL

Antistatic Flame-Retardant Shell & Heavy-Duty Center Zipper

Industrial grey antistatic and flame-retardant outer fabric with reinforced black trim and bilateral compression straps resists sparks, slag, and heavy abrasion.

ANATOMICAL & MECHANICAL SUBSYSTEMS (4 NODES)

1. **Flame-Retardant V-Neck Outer Shell [Antistatic FR V-Neck Armor Shell]**
   Minimalist grey antistatic shell with heavy-duty center zipper fits comfortably over base layers.

2. **Bilateral Compression Webbing [Bilateral Torso Compression Straps]**
   Side tension straps maximize thermal contact between the internal chilled TPU bladder and torso.

3. **Micro-Inverter Compressor Chiller [60-300W Micro-Inverter Refrigeration Core]**
   24V miniature compressor core stabilizes chilled water circulation within 3 minutes.

4. **Internal TPU Cooling Bladder & Insulated Loop [Pressurized Closed-Loop TPU Bladder]**
   Closed-loop liquid network paired with breathable 3D lining prevents condensation accumulation.

FIELD DEPLOYMENT | BLAST FURNACE & MOLTEN METAL CASTING BAY | >45°C AMBIENT



Extreme Metallurgy Foundries, Firefighting & Confined-Space Operations

In blast furnace casting bays, heavy equipment cabs without HVAC, and deep confined-space rescue missions, radiant heat loads overwhelm conventional cooling vests. Delivering 60-300W of active vapor-compression refrigeration in a <4kg wearable package, the LC Frost-ACC/W stabilizes chilled water within 3 minutes and protects core body temperature under the most punishing industrial conditions on earth.

60-300W (24V / 110W)	3 Minutes	< 4.0kg
Active Refrigeration Range	Chilled Loop Stabilization	Complete System Weight

COMPLETE FACTORY TECHNICAL SPECIFICATIONS · LC Frost-ACC/W (100% UNABBREVIATED)

ENGINEERING GROUP	PARAMETER	VERIFIED FACTORY SPECIFICATION
IDENTITY & COMPLIANCE	**Model**	LC Frost-ACC/W
IDENTITY & COMPLIANCE	**Color**	Industrial Grey with Black Trim
IDENTITY & COMPLIANCE	**Target Users**	Extreme high-temperature environments (metallurgy, foundry, firefighting, heavy equipment cabs)
PERFORMANCE & MECHANICS	**Cooling Capacity**	60-300W
PERFORMANCE & MECHANICS	**Rated Voltage**	24V
PERFORMANCE & MECHANICS	**Rated Power**	110W
PERFORMANCE & MECHANICS	**Cooling Response Time**	Stable in 3 min
PERFORMANCE & MECHANICS	**Battery Life**	1-2 hours single battery (2-4 hours system continuous operation)
MATERIALS & ERGONOMICS	**Outer Fabric**	Antistatic and flame-retardant fabric
MATERIALS & ERGONOMICS	**Lining Fabric**	High-durability TPU liquid cooling bladder
DIMENSIONS & WARRANTY	**Size**	One Size Fits All
DIMENSIONS & WARRANTY	**Net Weight**	< 4kg
DIMENSIONS & WARRANTY	**Package Dimensions**	40 x 30 x 10cm
DIMENSIONS & WARRANTY	**Warranty**	1-Year Official Industrial Warranty

DIRECT FACTORY PROCUREMENT & PILOT DEPLOYMENT - LC Frost-ACC/W

Manufacturer: Guangdong Langcheng Technology Co., Ltd. | R&D Base: Intelligent Protection Technology R&D Center (GDUT Institute of CNC Equipment Collaborative Innovation)

Hotline / WhatsApp: +86 159-8917-5854 | Engineering Email: 1094099100@qq.com | Warranty: 1-Year Official Industrial Manufacturer Warranty Across All Hardware